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Abstract

1 Introduction

Lake ice in the Great Lakes is highly variable and has impacts ranging from economy to winter ecology. The presence and timing of ice cover influences the length of the commercial shipping season and during high ice years shipping has been entirely suspended which interrupts supply chains that rely on raw materials such as Iron Ore, Grain, etc. (Each year 135 million tons of cargo are moved in the Great Lakes which support a quarter million jobs (Martin, n.d.)). Additionally, many coastal communities rely on ice cover for winter tourism which boosts local economies. [citation] The biota of the lakes interact with ice cover which impacts nutrient loading (Nicholls, 1998), serves as a winter habitat for Diatoms (Twiss et al., 2012) and has been widely suggested to protect Lake Whitefish eggs during otherwise harsh winter conditions (Brown et al., 1993; Collingsworth et al., 2017; Lynch et al., 2015).

The timing and extent of Great Lakes ice vary considerably from year to year. For example, the long-term (1973-2025) average and standard deviation for annual maximum ice cover is 52% and 24%, respectively, which implies that peak ice cover for a given year is frequently ($\mu \pm \sigma$) as low as 28% or as high as 76%. The length of the ice season is, on average, 82 days long (based on 10% threshold) but has been as short as 23 days and as long as 148 days. This variability has significant impact on Great Lakes ecosystems and further motivates the need to understand the drivers of ice cover in order to make meaningful predictions.

Previous studies have demonstrated that, consistent with trends in air and ocean temperatures, water temperatures in the Great Lakes, are also increasing both in the surface (Austin and Colman, 2007; O'Reilly et al., 2015; Woolway and Merchant, 2018; Zhong et al., 2018) and subsurface (Anderson et al., 2021; Cannon et al., 2024). However, the rate and extent of warming in the Great Lakes varies considerably from lake to lake and even *within* each lake (Mason et al., 2016), partly due to the heterogeneity of characteristics within lakes (bathymetry, latitude, surface area) often referred to as lake morphometry. These characteristics can act as competing factors for changes in lake ice cover. For example, Sharma et al. (2019) found that mean depth influenced ice loss for lakes at similar latitudes. Richardson et al. (2024) hypothesized that, globally, lake ice phenology will respond non-linearly to climate-change such that lakes in geographically colder (warmer) regions experience increased (decreased) interannual variability. This is mostly consistent with the findings of Lin et al. (2022), who showed that regionally the Great Lakes have been experiencing increased variability since 1998.

Assel (2005) classified ice seasons as either mild, typical or severe and found that. . .

Numerical ice forecasts are high resolution and go beyond the sub-regional scale but they are computationally expensive and have showed varied skill with some lakes and years more prone to error (Anderson

et al., 2018). Subregional ice forecasts Venumuddula et al., 2024 have proved advantageous over forecasts at larger scales.

To further understanding of the physical drivers of ice cover formation and decay, we analyze 54 years of daily ice cover in the Great Lakes at sub-lake scales. By evaluating ice cover aggregated into within subregions, we gain insight into changes and relationships occurring at the sub-lake scale that might otherwise be masked by those at the lake-wide scale. Furthermore, the heterogeneity of physical characteristics (e.g. depth) within the Lakes motivates an analysis of how those varying characteristics may be impacting long-term changes in ice cover. Additionally, we leverage coastal meteorological station data to assess how locally-specific atmospheric conditions impact the evolution ice cover over the course of a given ice season.

2 Methods

2.1 Overview

Following the work of Mason et al. (2016), we aggregate gridded ice cover data into 22 regions defined by the Great Lakes Aquatic Habitat Framework (L. Wang et al., 2015) (hereafter "subregions", see figure 1) which have been thoughtfully defined to delineate the Lakes (and watersheds) by various factors including circulation patterns, thermal stratification, as well as physiochemical and biological characteristics (see L. Wang et al., 2015 for more details.) Once aggregated, three ice metrics are calculated for each of the 54 ice seasons on record: annual maximum ice cover, seasonal average ice cover and ice duration. Though these ice metrics are closely correlated, each provides additional information about a given ice season.

The first component of analysis is identifying what statistically significant ice trends, if any, exist at these sub-lake scales, and how those changes might relate to the characteristics (e.g. morphometrics) of a given subregion. The second component addresses how atmospheric phenomena plays a role in dictating ice cover on the local scale. This is accomplished by using daily meteorological data from land-based stations specific to each subregion to identify correlations between atmospheric conditions and ice phenology. Variables analyzed include air temperature, wind speed, cloud cover, dew point (not shown), and precipitation.

By identifying subregional ice trends, we gain deeper insight into how lake ice is changing and what might be expected in the future at a more detailed level for each specific region. Additionally, relating these changes back to physical characteristics provides clues as to what drives the change and what makes a specific subregion or lake more susceptible to that change. These results can have implications for large lakes globally especially those at similar latitudes or with similar characteristics.

A analyze ice phenology, trends and relationships to both lake morphometrics and meteorological data. This analysis is described in detail next.

2.2 Metrics

2.3 Data

High-resolution ice cover in the Great Lakes has been recorded since 1973¹ coinciding with the advent of consistent satellite overpass in the region. This data set is made possible by the North American Ice Services (NAIS: comprised of US National Ice Center and Canadian Ice Service) who generate daily (semi-weekly prior to 2011) data using a variety of sources including remote-sensing data (synthetic-aperture radar, infrared and visible imagery), ship reports, ice reconnaissance flights and analyst knowledge. This data set was standardized by Yang et al. (2020) via re-sampling onto a 1.25 KM grid for the entire period of record and also temporally interpolated from semi-weekly to daily frequency. We use this standardized data set (hereafter: NAIS data) for all of our analysis.

¹Though some ice measurements pre-date 1973, they did not included comprehensive coverage of the lake surface so will not be considered in this work [(see CITE: TM-121 for details.)]



`../figures/station_map.png`

Figure 1: Figure caption

Meteorological data is available from the Great Lakes

While much previous work has identified the significant influence of teleconnections on (Great) Lakes ice cover (Bai et al., 2012; Richardson et al., 2024; J. Wang et al., 2018), our focus is on subregional ice formation (i.e. differences between

- hange point analysis for each region?
- should we subset recent data to match frequency of older data? or should we repeate analysis using the daily interpolated data?
- include model data (or mooring data) for subregions stratification or heat content? this is a biiig can of worms
- mooring data might be doable but limited regions and years

3 Results

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